Chemistry

1 The insoluble product formed is Barium sulphate [$BaSO_4$] and color is white

$$\underset{(aq)}{Na_2SO_4} + \underset{(aq)}{BaCl_2} \longrightarrow \underset{(s)}{BaSO_4} + \underset{(aq)}{2NaCl}$$

2 Ammonia gas

3 Both assertion & reason are true. Reason is correct explanation of Assertion

4 Assertion is true but reason is false.

5 The amphoteric oxides ~~are~~ are those which react ~~wh~~ with both acids as well as base in order to form salt and water.

~~Amphoteric oxides behave like~~

Examples: Zinc oxide (ZnO), Aluminium oxide (Al_2O_3)

Balanced Equation:-

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \longrightarrow \underset{\text{(sodium aluminate)}}{2NaAlO_2} + H_2O$$

6 The student will observe displacement reaction in beakers having Aluminium and Zinc strips

7 On heating green vitriol crystals ($FeSO_4 \cdot 7H_2O$) ~~following the reaction~~ to thermal decomposition takes place

$$2FeSO_4 \longrightarrow \underset{(\text{white})}{Fe_2O_3} + SO_2 + SO_3$$

The colour changes from light green to white
and due to formation of SO_2 and SO_3 pungent smells come

8 (a) Zinc granules

$$\underset{\substack{(s)\\ \text{Zinc}}}{Zn} + \underset{\substack{(aq)\\ \text{Hydrochloric}\\ \text{acid}}}{2HCl} \longrightarrow \underset{\substack{(g)\\ \text{hydrogen}\\ \text{gas}}}{H_2} + \underset{\substack{(aq)\\ \text{zinc}\\ \text{chloride}}}{ZnCl_2}$$

(b) Crushed egg shells

$$\underset{\substack{\text{Hydrochloric}\\ \text{acid}}}{2HCl} + \underset{\substack{\text{Calcium}\\ \text{carbonate}}}{CaCO_3} \longrightarrow \underset{\substack{\text{Calcium}\\ \text{chlo}\\ \text{chloride}}}{CaCl_2} + \underset{\substack{\text{Carbon}\\ \text{dioxide}}}{CO_2} + \underset{\text{water}}{H_2O}$$

9 The salt used is Baking soda ($NaHCO_3$)

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

Baking soda is produced when excess CO_2 is passed from saturated solution of sodium carbonate

Uses = ① soda acid fire extinguisher
② To make baking powder and to make antacids

10

(b

10 (a) Chemical equation involved in preparation of sodium hydroxide (NaOH).

$$\underset{(aq)}{2NaCl} + \underset{(l)}{2H_2O} \xrightarrow{electricity} \underset{(aq)}{2NaOH} + \underset{(g)}{Cl_2} + \underset{(g)}{H_2}$$

The process is called chlor-alkali process.

(b) When CO_2 is passed through lime water [$Ca(OH)_2$] it turns milky due to formation of calcium carbonate ($CaCO_3$)

$$Ca(OH)_2 + CO_2 \longrightarrow \underset{\text{(insoluble in water)}}{CaCO_3} + H_2O$$

When excess CO_2 is passed, milkiness disappears due to formation of calcium bicarbonate [$Ca(HCO_3)_2$] which is colourless & soluble in water.

$$CaCO_3 + H_2O + \underset{\text{(in excess amount)}}{CO_2} \longrightarrow Ca(HCO_3)_2$$

11 (i) $$\underset{(s)}{CaO} + \underset{(s)}{SiO_2} \longrightarrow \underset{(s)}{CaSiO_3}$$

- This equation is already balanced. Type of chemical reaction: combination reaction

② $TiCl_4$ (l) + $MgCO_3$ (s) ⟶ Ti (s) + $MgCl_2$ (s)

Balanced : $TiCl_4$ (l) + $2Mg$ (s) ⟶ Ti (s) + $2MgCl_2$ (s)

Type : Displacement reaction

③ H_2O_2 (l) $\xrightarrow{U.V.}$ H_2O (l) + O_2 (g)

Balanced : $2H_2O_2$ (l) $\xrightarrow{U.V}$ $2H_2O$ (l) + O_2 (g)

Type : Decomposition reaction

④ $Pb_3O_4 + HCl \longrightarrow PbCl_2 + Cl_2 + H_2O$

Balanced : $Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl$

Type : Redox reaction

⑤ $KCl + AgNO_3 \longrightarrow AgCl + KNO_3$

Balanced : It is already balanced !

Type : Double displacement reaction or precipitation reaction